Ordering the Evolution and Function of Intrinsically Disordered Regions in Neurodegenerative Proteins

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ABSTRACT

This research employs computational approaches to investigate the evolutionary trajectories of IDRs in TDP-43, FUS, and Axin-1, proteins implicated in NDs. By identifying residue-level evolutionary adaptations within IDRs, this study aims to bridge gaps in functional understanding and uncover potential causes of human neurodegenerative susceptibility. Using *Homo sapiens* as a reference, alongside *Bos taurus*, *Mus musculus*, *Pan troglodytes* and *Xenopus laevis*, this work spans different phylogenetic distances to identify conserved and divergent patterns in IDR evolution. These species were chosen to capture both closely related and distantly related evolutionary contexts, enabling the identification of human-specific IDR adaptations and conserved functional motifs. Unknown IDRs will be predicted and labeled using AlphaFold-Disorder, while functional residue annotations will be derived from the Eukaryotic Linear Motifs (ELM) database. Multiple sequence alignment (MSA) and phylogenic tree construction are used to identify evolutionary divergence and convergence among IDRs of selected species. To further explore functional features, residue embeddings will be generated using ProtTrans, a large language model trained on structural and functional protein data. These embeddings, combined with ELM annotations, will be analyzed using unsupervised learning techniques such as t-distributed stochastic neighbor embedding (t-SNE) and density-based spatial clustering of applications with noise (DBSCAN). By clustering residues based on ProtTrans-defined features and surveying ELM label enrichment, this study will reveal how residues with similar characteristics exhibit diverse functional roles across species. This work aims to provide an evolutionary perspective on IDR function, offering insights into the biological underpinnings of human neurodegenerative susceptibility and contributing to the broader understanding of disordered protein evolution and function.

Keywords: Intrinsically disordered regions (IDRs), ProtTrans, AlphaFold, protein evolution, t-SNE, phylogenic analysis, machine learning

Project type: Research

Project repository: https://github.com/gpadill5/BIOCB_6840_final

1 Introduction

Intrinsically disordered regions (IDRs) are regions within proteins that lack secondary structures, contributing to their unique functionality and involvement in neurodegenerative diseases (NDs) such as Alzheimer's disease, Parkinson's disease (PD), and amyotrophic lateral sclerosis (ALS) [1]. These disordered regions enable proteins to perform numerous biological functions, often involved in cellular signaling and aggregation, which are critical to ND pathology [1]. Proteins such as TDP-43, FUS, and Axin-1 contain IDRs that are central to ALS and PD.

TDP-43 and FUS are DNA/RNA-binding proteins involved in RNA metabolism, including transcription regulation, RNA splicing, and transport [1, 2]. TDP-43 malfunction in ALS is associated with the cytoplasmic aggregation of post-translationally modified forms of the protein, including phosphorylation, ubiquitination, and truncations [2]. Similarly, FUS forms cytoplasmic aggregates in ALS, although these tend to have fewer post-translational modifications than TDP-43 aggregates, they still contribute to the ALS disease phenotype. In both cases, IDRs are the main driver of aberrant cytoplasmic aggregation and post-translational modifications in TDP-43 and FUS [1, 2]. Axin-1, on the other hand, is a regulator in the Wnt/β-catenin signaling pathway, a pathway implicated in neurodegeneration and brain development [3]. In PD, Axin-1 expression is upregulated, indicating its potential involvement in PD progression through dis-regulation of the Wnt/β-catenin pathway. Axin-1's IDR provides structural flexibility, enabling promiscuous binding and signaling within its environment, which may facilitate a toxic gain-of-function under pathological conditions [3].

While proteins like TDP-43, FUS and Axin-1 are not human-specific proteins, NDs for which such proteins are pathologically responsible predominantly affect humans as they age. This suggests that the prolonged human lifespan contributes to a higher susceptibility to NDs, raising significant concerns for our aging society. Evolutionary constraints likely do not strongly select against these conditions, as NDs manifest post-reproductively in most cases, allowing for unconstrained gene flow [4]. Because evolutionary constraints do not strongly select against deleterious IDR mutations there is greater divergence in IDR-containing proteins across species. This results in a lack of conservation and species-specific variations in splicing and isoforms, further compounded by incomplete experimental and annotated datasets [4, 5]. For these reasons, understanding the evolutionary relationships of IDR-containing proteins across species is challenging but essential.

To address these problems, this project compares the IDRs of TDP-43, FUS, and Axin-1 across five species—*Pan troglodytes* (chimpanzee), *Mus musculus* (mouse), *Bos taurus* (bovine), and *Xenopus laevis* (African clawed frog)—with *Homo sapiens* serving as the referential species. This selection spans a range of phylogenetic distances, offering insights into conserved and species-specific patterns in IDR structure and function. By using humans as a reference, the study highlights unique IDR adaptations that may contribute to neurodegenerative disease susceptibility. Comparisons with closely related species, such as chimpanzees, can help identify subtle variations, while comparisons with distantly related species, such as frogs, provide a broader evolutionary window for identifying conserved motifs crucial to protein function. This approach enables a nuanced understanding of how IDRs have evolved to meet functional requirements and their roles in human NDs.

To investigate the evolutionary and functional characteristics of IDRs in TDP-43, FUS, and Axin-1 between multiple species, I begin by searching for annotated IDR sequences in the respective protein entries using MobiDB. For species without annotated IDRs, I predict their residue labeling using the AlphaFold-Disorder tool, which uses structural predictions from the AlphaFold database as input. This method identifies disordered regions by correlating residue solvent accessibility (RSA) in predicted protein structures to IDRs [6]. Once the IDR sequence ranges for each protein are established, I perform a multiple sequence alignment (MSA) to align IDR sequences across species and generate a phylogenetic tree. MSA allows for the identification of conserved and non-conserved residues, providing an evolutionary context for further data processing.

For functional analysis, I focus on non-conserved residues filtered through MSA to uncover divergent functional roles across different species. High-dimensional residue embeddings were generated using ProtTrans, a protein-based large language model that captures sequence-aware residue features using self-supervised learning. ProtTrans enables the analysis of IDRs by embedding amino acid sequences into high-dimensional spaces, revealing structural and functional protein data beyond MSA-based approaches [7]. These embeddings are then integrated with residue-specific labels from the Eukaryotic Linear Motifs (ELM) database, a catalog of short linear motifs essential for protein function and interaction networks [8]. By applying unsupervised learning techniques, such as t-distributed stochastic neighbor embedding (t-SNE) for dimensionality reduction and density-based spatial clustering of applications with noise (DBSCAN) for clustering, I assess ELM label enrichment within clusters to approximately map ELM-derived function to ProtTrans embedding space. This approach identifies functional deviations in *Homo sapiens* IDRs, revealing unique adaptations and potential links to neurodegenerative disease susceptibility.

2 Methods

Data Collection Protein sequences and IDR annotations were collected from publicly available databases, including UniProt [9], MobiDB [10], and DisProt [11]. These databases provide comprehensive repositories of experimentally validated and computationally predicted protein disorder data, ensuring high-quality and diverse input datasets. For proteins lacking IDR annotations, structures were sourced from the AlphaFold Protein Structure Database [12], leveraging its accurate structural predictions for downstream IDR prediction and labeling.

To investigate evolutionary and functional patterns, five species (*Homo sapiens, Pan troglodytes, Mus musculus, Bos taurus,* and *Xenopus laevis*) were chosen to represent a range of evolutionary distances, from closely related species (*H. sapiens* and *P. troglodytes*) to more distantly related species (*X. laevis*). This selection allows for the identification of conserved and species-specific patterns in IDR structure and function, with mammals (*H. sapiens, P. troglodytes, M. musculus, B. taurus*) providing insights into vertebrate-specific adaptations and *X. laevis* offering a broader phylogenetic perspective. TDP-43, FUS, and Axin-1 were selected as proteins for this study due to their vital roles in human NDs and their functional dependence on IDRs.

Intrinsically Disordered Region Labeling Using AlphaFold-Disorder Tool IDRs were identified using the AlphaFold-Disorder tool, which utilizes AlphaFold predictions, particularly relative solvent accessibility (RSA) and predicted Local Distance Difference Test (pLDDT) output values, as proxies for structural flexibility. An RSA threshold of 0.581 was selected based on optimized performance in predicting IDRs [6]. RSA was prioritized over pLDDT due to its higher accuracy in distinguishing disordered regions from secondary structures (e.g., α-helices, β-sheets) [6].

To reduce noise, 1D Gaussian filtering was applied to smooth RSA values, using a standard deviation (σ) of 7. This parameter was chosen iteratively to maximize agreement between predicted and experimentally validated IDRs, measured by the Jaccard index (Supplementary Fig. 1). Regions where smoothed RSA values exceeded the 0.581 threshold were classified as disordered, while those below the threshold were classified as ordered. These classifications were visualized on domain maps, with IDRs depicted as red bands and ordered regions as blue bands (Fig. 1). Smoothed-RSA predicted IDR labels were validated using MobiDB-labeled *Homo sapiens* IDRs as a reference. This method provides a robust and lightweight framework for identifying IDRs in proteins where such annotations are lacking.

Multiple Sequence Alignment and Phylogeny Tree Construction Multiple sequence alignments of IDR sequences were performed using Clustal Omega to identify conserved residues and patterns of evolutionary divergence [13] (Supplementary Fig. 2, 3, 4). Pairwise distances were calculated to construct a guide tree using the UPGMA algorithm, followed by progressive alignment and merging. A phylogenetic tree was inferred using the neighbor-joining algorithm to map the evolutionary trajectories of IDRs between species (Fig. 2).

ProtTrans-Based Embedding and Clustering for Functional Analysis of IDRs Residue embeddings were generated using ProtTrans, a large protein-based language model from the Hugging Face `transformers` library. ProtTrans embeddings capture sequence-aware features in a 1024-dimensional space, representing structural and functional properties of residues [7].

Clusters of non-conserved residues, identified through MSA, were embedded using ProtTrans to capture their distinct features. Functional motifs were annotated using the ELM database, a resource cataloging short linear motifs involved in protein functions such as ligand-binding and modification sites [8]. These annotations were integrated with ProtTrans embeddings to map functional enrichment onto the embedding space.

Clustering was performed using DBSCAN, which identifies dense regions in the embedding space while effectively handling noise. DBSCAN was chosen for its ability to cluster data points based on density, as ProtTrans embeddings are non-linear and high-dimensional. Clusters were subsequently labeled according to the dominant ELM label enrichment within each group, providing a proxy for assigning functional categories to the ProtTrans embedding space. This method facilitated the association of biologically meaningful functional labels with high-dimensional embeddings, revealing species-specific adaptations in IDR functionality.

Relative Enrichment Analysis of ELM Labels Across Selected Species Clusters enriched with specific ELM annotations were labeled and visualized using an intensity color bar, where higher vibrancy indicated greater enrichment (Fig. 3). The relative percentage of each enriched cluster group was calculated for each species as:

$$\text{ELM Label Enrichment (\%)} = \frac{\text{Count of Residues for ELM Label}}{\text{Total Residues for Species}} \times 100$$

Grouped histograms were constructed to visualize enrichment across species (Fig. 4). Statistical significance was assessed using a two-proportion Z-test for each label, comparing proportions of ELM-labeled residues in humans versus other species, with a significance level of $\alpha = 0.2 (p < 0.2)$. A significance level of $\alpha = 0.2$ was chosen to balance sensitivity and the small dataset size, ensuring meaningful detection of trends while minimizing false negatives

3 Results

Smoothed RSA Accurately Predicts IDRs in Unannotated Sequences IDRs were successfully predicted for unannotated sequences using the AlphaFold-Disorder tool, which utilizes RSA as an indicator of disorder within predicted protein structures. Smoothed and filtered RSA ranges were utilized to map predicted disordered regions in TDP-43, FUS, and Axin-1 proteins for which annotations for some species were unavailable in MobiDB (e.g., TDP-43 *X. laevis*) (Fig. 1. The resulting domain maps show good agreement with known, MobiDB-sourced IDR positions in MSA (Supplementary Fig. 2, 3, 4), supporting the accuracy of smoothed RSA-based IDR prediction.

Conservation Patterns in Axin-1 IDRs The MSA of Axin-1 reveals high conservation of IDR 1 among mammals, including *Homo sapiens* (HUMAN), *Pan troglodytes* (CHIMP), *Mus musculus* (MOUSE), and *Bos taurus* (BOVIN), while *Xenopus laevis* (XENLA) displays significant divergence, especially in IDRs 2 and 3 (Supplementary Fig. 4). The phylogenetic tree (Figure 2, A) roughly clusters together mammalian species demonstrating minor divergence, while positioning X. laevis at a basal branch (0.159739 branch length), which agrees with its reduced IDR conservation. This divergence can be seen in Fig. 4, C, where *X. laevis* exhibits the highest percentage of modification site residue labeling within the modification-enriched cluster by 10%.

High Conservation in Fus IDRs The MSA of FUS reveals strong conservation of intrinsic disordered regions (IDRs) across all species analyzed. IDR sequences remain nearly identical among mammals, with minor sequence variations in IDRs observed in *B. taurus* and *X. laevis* (Supplementary Fig. 3). The phylogenetic tree (Fig. 2, B, confirms this conservation through its branch lengths. Short branch lengths connecting mammals display their recent divergence, with lengths 0.004 0.002. In contrast, *X. laevis* with the longest branch, with a length of 0.136 reflects its earlier divergence from mammals. Despite this however, *X. laevis* retains conserved motifs within FUS IDRs.

Intermediate Conservation in TDP-43 IDRs The MSA of TDP-43 also reveals high conservation of intrinsic disordered regions (IDRs) across mammals, with divergence observed in X. laevis (Supplementary Fig. 2). Similar to FUS, mammalian species share near-identical IDR sequences, consistent with the short branch lengths in the phylogenetic tree (Fig. 2, A) with branch lengths on the order of 0.01 . In contrast, *X. laevis* had the longest branch within the tree (0.08), indicative of greater evolutionary distance.

Functional Divergence of IDRs in Humans Compared to Selected Species Residue-level embeddings of non-conserved residues within IDRs were generated using ProtTrans. These embeddings were visualized in a reduced-dimensional space with t-SNE (Figure 3), which was selected for its effectiveness in preserving data relationships within high-dimensional data. Clustering with DBSCAN identified functionally distinct regions within the embedding space. These clusters were annotated with Eukaryotic Linear Motif (ELM) labels to assign biological function. For all three proteins—TDP-43, FUS, and Axin-1—DBSCAN clusters were inspected for ELM annotation enrichment and re-labeled to match the ELM label majority for a given cluster, such as ligand-binding and residue modification, to approximate embedding space-function relationship.

Grouped histograms of ELM label enrichment across species (Figure 4) reveal significant functional differences in *Homo sapiens* IDRs compared with other species. For TDP-43, *Homo sapiens* showed a comparatively significant enrichment (p < 0.2) in modification site labeled residues where residues similar in embedding space are predominantly ligand-binding sites. Additionally, within TDP-43 modification site embedding space, *Homo sapiens* has one of the highest percentages of residues that accurately match modification site labeling. In FUS,*Homo sapiens* showed no noticeable difference in functional labeling within enriched clusters. However, *Homo sapiens* displays variability in Axin-1 residue function compared with other species. In Axin-1's ligand-binding enriched embedding area, *Homo sapiens* Axin-1 exhibits lower targeting signals than expected. Conversely, in Axin-1's targeting signals-enriched embedding area, *Homo sapiens* exhibits a higher prevalence of docking motifs compared to other species, whereas its modification site enrichment is relatively low. These findings underscore human-specific adaptations in IDRs, potentially linked to neurodegenerative disease susceptibility.

3.1 Synthetic Data

AlphaFold Structure Database, AlphaFold-Disorder tool.

3.2 Real Data

UniProt Knowledgebase (UniProtKB), DisProt Database, MobiDB Database, Eukaryote Linear Motif (ELM) Database

4 Discussion

This study explored the evolutionary conservation and functional divergence of IDRs in TDP-43, FUS, and Axin-1 across five species. By combining computational approaches—including AlphaFold-Disorder, multiple sequence alignments (MSAs), phylogenetic analysis, and ProtTrans embeddings analyzed with unsupervised learning techniques like DBSCAN—I uncovered patterns of conservation, divergence, and enrichment of functional labels. The study's results showed high conservation of IDRs within mammals, but significant divergence in *X. laevis*, particularly for Axin-1. Furthermore, functional clustering may hint at human-specific adaptations in IDRs that contribute to human ND susceptibility.

The evolutionary analysis of TDP-43, FUS and Axin-1 IDRs provided distinct patterns of conservation and divergence across species. In TDP-43, the high conservation of core IDR motfis across mammals highlight their critical roles in RNA-binding and stress granule formation [2]. However, the divergence observed in the IDR termini of *X. laevis* (shown in Supplementary Fig. 2) potentially suggesting species-dependent variations in RNA metabolism. Similarly, FUS exhibited strong conservation of IDRs among mammals, with *X. laevis* conserving most residues despite its earlier evolutionary divergence (Fig. 2). This indicates that FUS's regulatory functions are under strong evolutionary pressure, even in distant taxa. This strong conservation of FUS may provide further insight into why FUS seems to be less neurotoxic than TDP-43 in *Homo sapiens* [1, 2]. In contrast, Axin-1 showed significant divergence in IDRs 2 and 3 in *X. laevis* (Supplementary Fig. 4), coupled with the highest enrichment of modification sites (Fig. 4, C). These differences may reflect different evolutionary pressures influencing Axin-1's signaling role in Wnt/β-catenin signaling pathways [3].

In addition to investigating inter-species evolutionary trends of IDRs, human-specific functional adaptions were also explored to potentially explain human ND susceptibility. In TDP-43, the enrichment of modification sites within ligand-binding clusters in humans (Fig. 4, A) supports the hypothesis that PTMs may help drive neurotoxicity in ALS TDP-43 aggregation. These findings align with preexisting literature that PTMs regulate TDP-43's aggregation and dysfunction [1, 2]. In Axin-1, the high levels of docking motifs labeling and lower levels of modification site labeling in targeting signal clusters may suggest differential regulation of Axin-1 in humans. Such adaptions may provide insight into Axin-1 function in human neuronal development and Parkinson's.

While this study underscores the strength of using self-supervised and unsupervised learning methods to transform raw biological data, in this case protein sequences, into biologically relevant insights, it does have its limitations. The scope of this study has been restricted to three proteins and five species, which not sufficient to accurately map out IDR evolutionary patterns. Additionally, the statistical threshold used ($p < 0.2$) is relatively weak, considering the data size of this work, larger data sets should yield a stricter statistical threshold. Moreover, quantitative methods to directly connect computational predictions to biological function is needed.

Future research should work to address these weaknesses by including a larger range of IDR-containing proteins and species to accurately map IDR evolution. Experimental validation through genetic editing (e.g., CRIPSR) and functional assays to validate computational predictions and provide mechanistic insight to the role of IDRs in NDs. Further, refining the statistical rigor of the enrichment work flow will enhance confidence in functional annotations and trends.

In summary, this study provides a framework for understanding the evolutionary trajectories of IDRs and their implications for human-specific vulnerabilities to NDs. By integrating computational and experimental approaches, future studies can build on these findings to uncover the complex relationships between protein disorder, evolution, and function, ultimately contributing to advancements in understanding IDR evolution.

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Figures

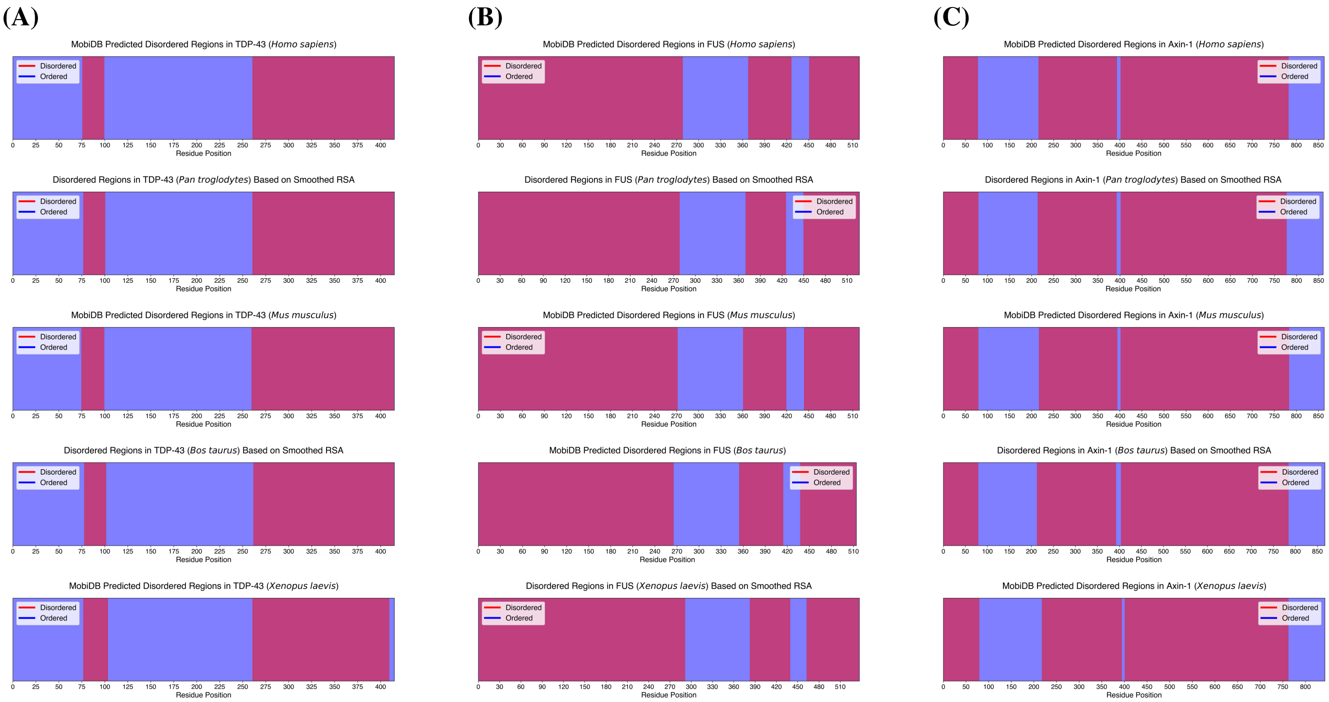

Figure 1. Domain maps of ordered and IDRs within TDP-43, FUS, and Axin-1 sequences (MobiDB-annotated and Alphafold-disorder derived). (A) TDP-43 intrinsically disordered regions (IDRs) and ordered regions for *Homo sapiens, Pan troglodytes, Mus musculus, Bos taurus, Xenopus laevis*. IDR annotations for *H. sapiens, P. troglodytes*, and *M. musculus* were sourced from MobiDB, while annotations for *X. laevis* and *B. taurus* were derived using the Alphafold-disorder tool and 1-D Gaussian smoothing. **(B)** FUS IDRs and ordered regions for *Homo sapiens, Pan troglodytes, Mus musculus, Bos taurus, Xenopus laevis*. IDR annotations for *H. sapiens, B. taurus*, and *M. musculus* were sourced from MobiDB, while annotations for *X. laevis* and *P. troglodytes* were derived using the Alphafold-disorder tool and 1-D Gaussian smoothing. **(C)** Axin-1 IDRs and ordered regions for *Homo sapiens, Pan troglodytes, Mus musculus, Bos taurus, Xenopus laevis*. IDR annotations for *H. sapiens, X. laevis*, and *M. musculus* were sourced from MobiDB, while annotations for *P. troglodytes* and *B. taurus* were derived using the Alphafold-disorder tool and 1-D Gaussian smoothing.

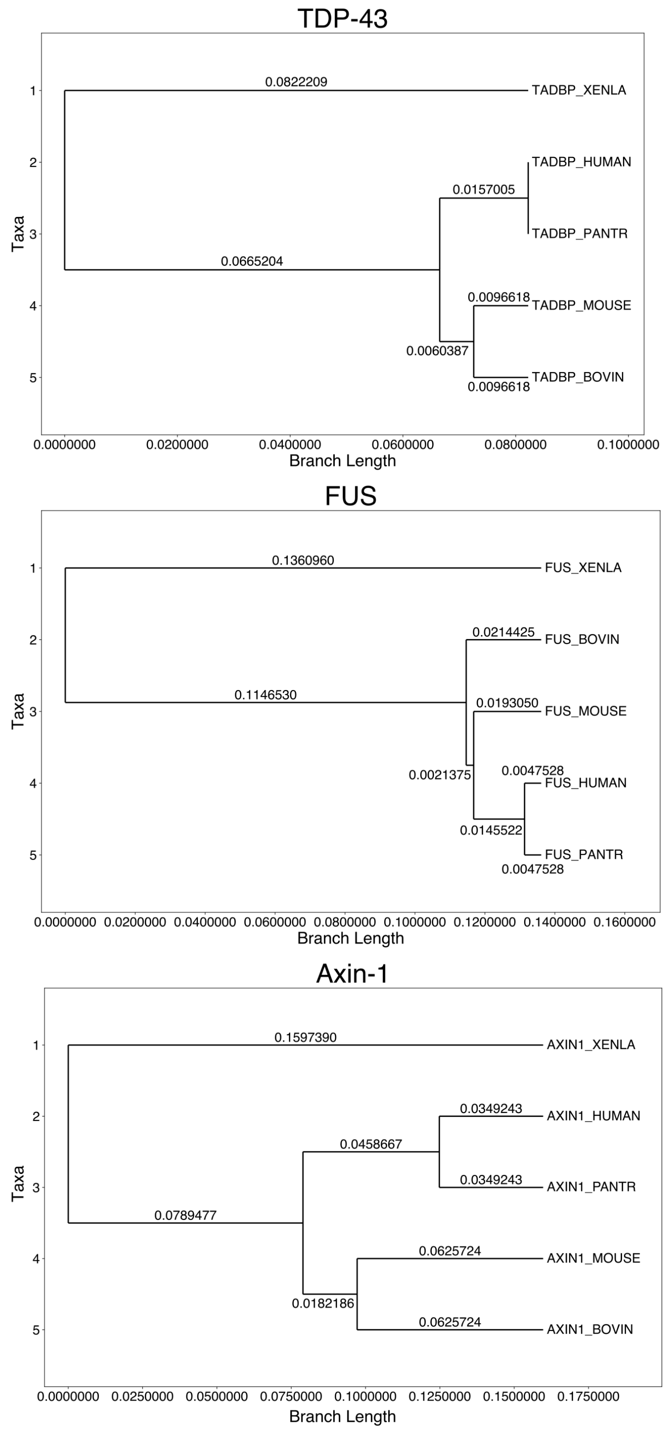

Figure 2. Phylogenetic trees of TDP-43, FUS, and Axin-1 proteins across selected species. The trees depict evolutionary relationships and branch lengths for the taxa *Xenopus laevis* (XENLA), *Bos taurus* (BOVIN), *Mus musculus* (MOUSE), *Homo sapiens* (HUMAN), and *Pan troglodytes* (PANTR). Branch lengths are labeled to indicate genetic distances.

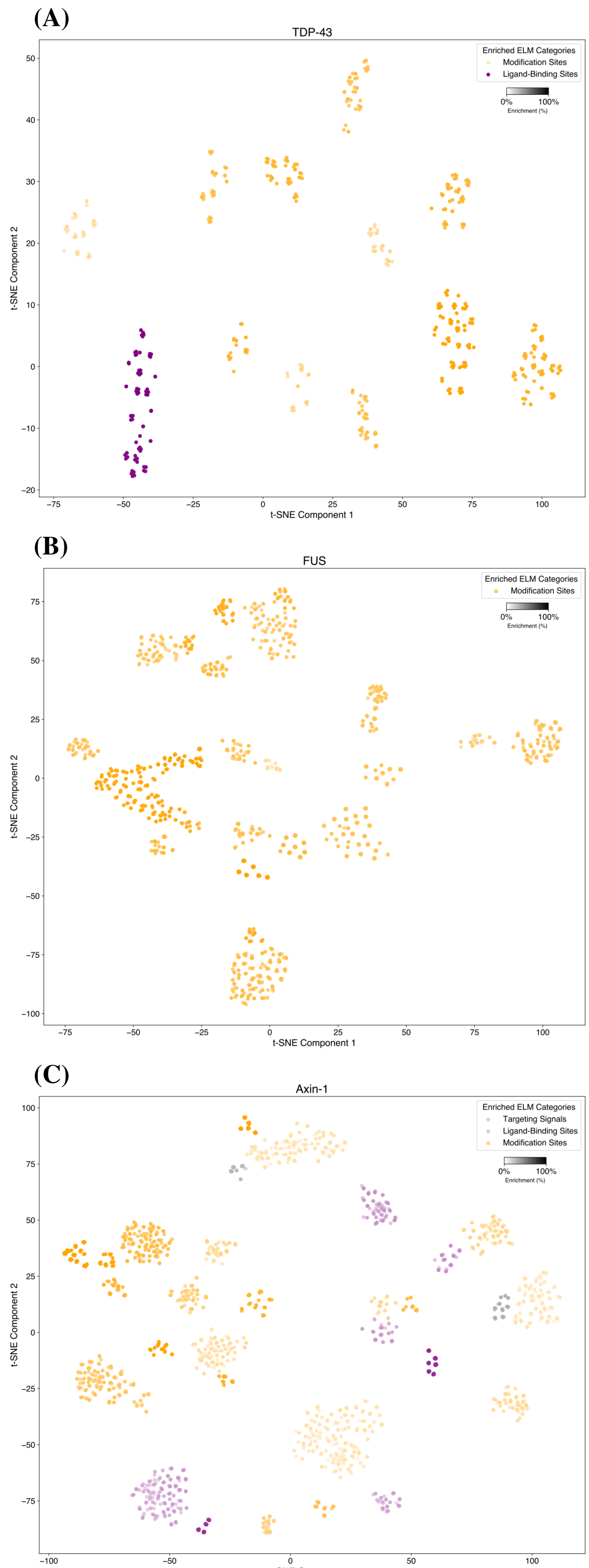

Figure 3. t-SNE visualizations of TDP-43, FUS, and Axin-1 IDR non-conserved residue embeddings and eukaryotic linear motifs across selected species. Sequences were analyzed to define intrinsically disordered regions (IDRs), which were aligned via multiple sequence alignment (MSA) to identify non-conserved residues within IDRs. ProtTrans was used to generate residue-level embeddings for these regions, while eukaryotic linear motifs (ELMs) were annotated using the ELM resource. The embeddings were reduced to two dimensions with t-SNE and clustered using DBSCAN. Clusters were assessed for enrichment in ELM annotations, allowing biological functions to be assigned within the ProtTrans embedding space. **(A)** t-SNE visualization of non-conserved residues from TDP-43's two IDRs, with clusters enriched for residue modification and ligand-binding site ELM categories. **(B)** t-SNE visualization of non-conserved residues from FUS's three IDRs, highlighting clusters enriched for residue modification-related ELMs. **(C)** t-SNE visualization of non-conserved residues from Axin-1's three IDRs, showing clusters with enrichment in signaling, ligand-binding and residue modification-related ELMs.

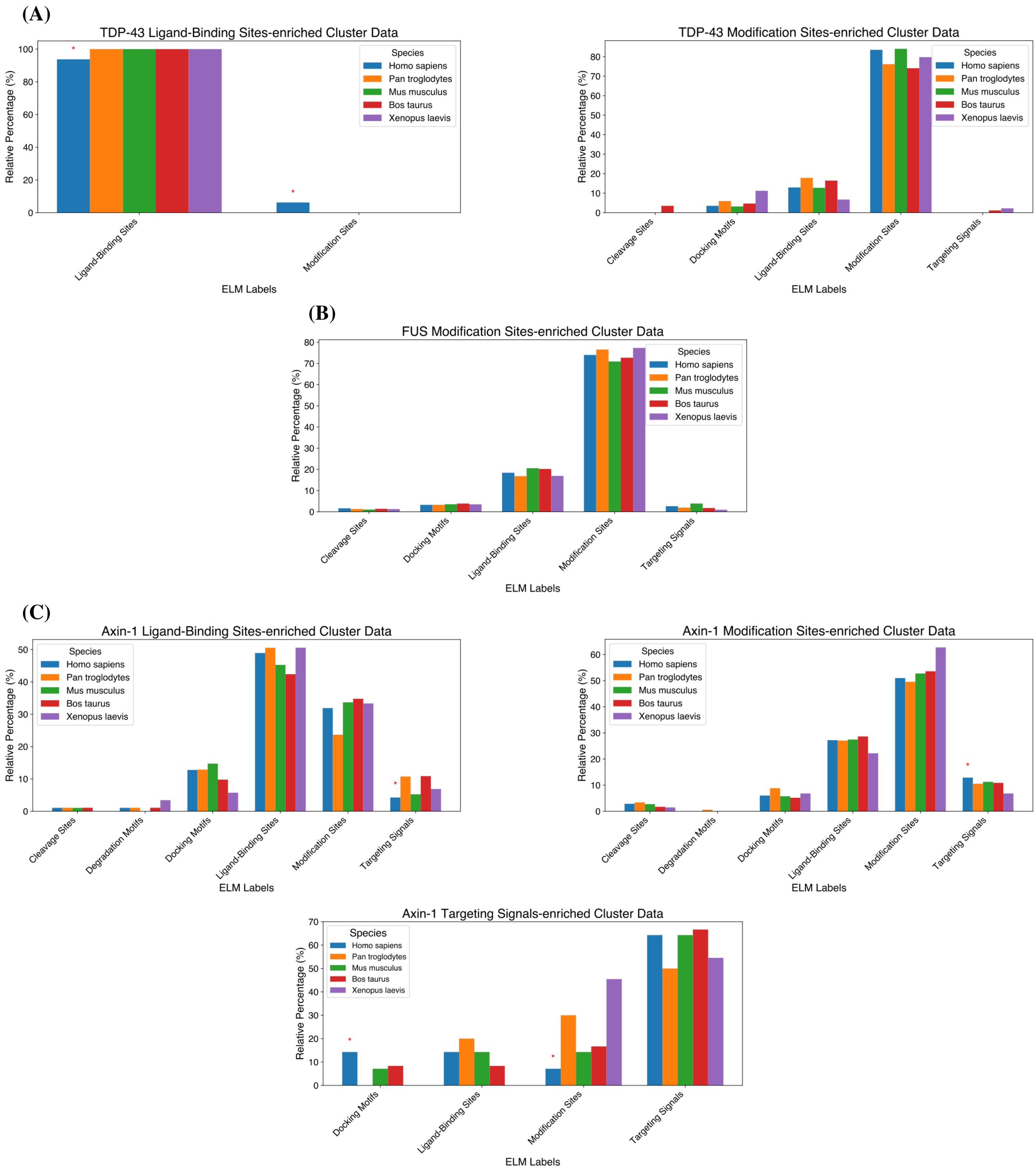

Figure 4. ELM label-enriched cluster data suggest functional divergence in *H. sapiens* IDRs. Grouped histograms depict the relative percentages of eukaryotic linear motif (ELM) labels for ELM label-enriched clusters of TDP-43, FUS, and Axin-1 non-conserved residue embeddings across species. Relative percentages are calculated for each species within the respective ELM label categories. Significance of enrichment for *H. sapiens* compared to all other species was assessed using a two-sided Z-test for proportions, with significant results ($p < 0.2$) marked by red asterisks. **(A)** TDP-43 ligand-binding site-enriched cluster data and modification site-enriched cluster data. (**B**) FUS modification site-enriched cluster data. (**C**) Axin-1 ligand-binding site-enriched cluster data, modification site-enriched cluster data, and targeting signals-enriched cluster data.

Supplemental Materials: Ordering the Evolution and Function of Intrinsically Disordered Regions in Neurodegenerative Proteins

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1 Section 1

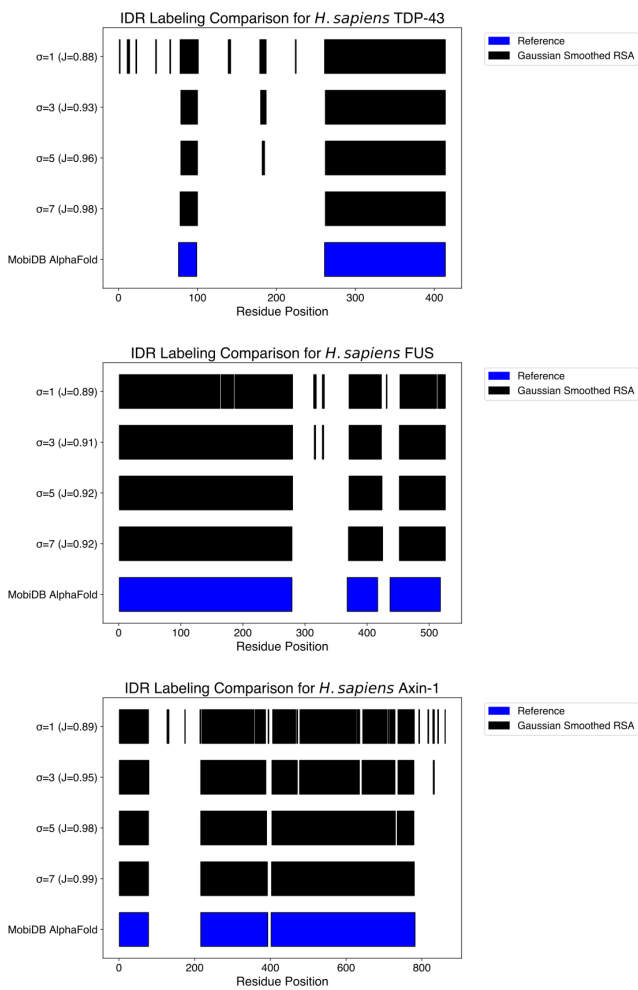

Figure 1. Comparison of IDR Labeling Between MobiDB AlphaFold Annotations and Gaussian-Smoothed AlphaFold-Disorder Predictions. Comparison of intrinsically disordered region (IDR) annotations from MobiDB AlphaFold (blue) and Gaussian-smoothed predictions from the AlphaFold-Disorder tool (black) across varying smoothing parameters ($\sigma = 1,3,5,7$). Residue positions are plotted as horizontal bars, with Jaccard indices quantifying the overlap between the two labeling, as a means of determining accuracy.

Figure 2. **MSA of TDP-43, FUS, and Axin-1 proteins across selected species.** Multiple sequence alignment (MSA) produced by Omega Clustal confirms IDR AlphaFold-Disorder labeling. Faded boxes represent residues outside of intrinsically disordered regions (IDRs), while solid colors indicate residues within IDRs. A solid line marks the boundary between disordered and ordered regions. Consensus notation is used to indicate sequence conservation: an asterisk ($\star$) denotes fully conserved residues, a colon (`:`) indicates strongly conserved residues with similar properties, and a period (`.`) represents weak conservation. Gaps in alignment are indicated by dashes ($-$) *(For MSA data (size limits), please refer to assignment attachments.)*

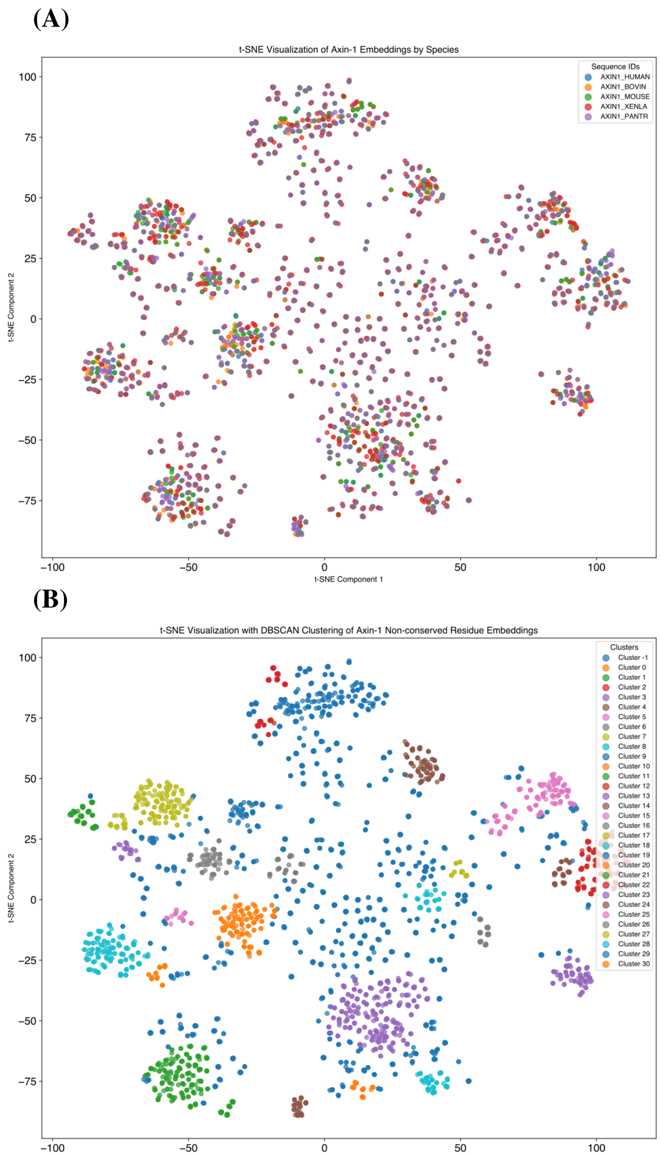

Figure 3. **t-SNE visualizations of Axin-1 IDR non-conserved residue embeddings across selected species.** **(A)** t-SNE visualization of from Axin-1's three IDRs, with residue data points labeled according to their respective species. **(B)** t-SNE visualization of Axin-1's three IDRs, annotated with clusters and noise identified using the DBSCAN algorithm from scikit-learn.